Candidate-Bound Evidence for RTL-to-GDS Signoff: An Open-Source Orchestration Study

Rithwik Vallabhan TV, Sreejeesh SG, Jayaraj U Kidav, and Nanditha N Varma

NIELIT Calicut, India
rithwik@calicut.nielit.in, sreejeesh@nielit.gov.in
jayaraj@nielit.gov.in, nanditha@calicut.nielit.in

Abstract. A clean physical-verification report is useful only if its checks cover the implementation selected for release. RTL2GDSAGI binds RTL, constraints, layout, parasitics, verification inputs, and native gate records to a candidate manifest and revalidates these identities before local signoff. We evaluate 7 SKY130 designs spanning 16–1024 mapped cells and complete the declared 21-stage contract, including synthesis and routed equivalence, extracted-parasitic timing, DRC, LVS, and antenna checks. A native fault-injection study covers six stale or incomplete evidence classes, with three replicates each. A comparator based on successful exit or cached status accepts all 18 fixtures; the candidate-aware checks block all 18 cases. Nine live model-assisted repair trials cover spacing, power-network construction, and stressed setup timing; 7 complete full rechecking and 2 escalate. At an unchanged 3 ns period, one ALU repair improves setup slack from -0.326192 ns to 0.001586 ns. Repeated implementation runs produce 6 identical routed-DEF pairs and 6 identical GDS pairs after zeroing timestamp payloads; raw GDS hashes and candidate identifiers can differ. All tables are generated from retained native artifacts and audit logs.

Keywords: RTL-to-GDS · physical verification · artifact provenance · electronic design automation · verified repair

1 Introduction

OpenROAD and OpenLane provide practical open infrastructure for synthesis and physical implementation [1,3]. Their native checks answer different questions: timing concerns a constrained circuit and parasitic network; DRC concerns geometry under a rule deck; and LVS concerns the layout and its schematic reference. A controller must establish that these answers support the same release candidate. Zero DRC markers can accompany a stale layout or an incomplete rule inventory, and a success message can appear without a real LVS comparison. Repairs create an additional opportunity for previously valid evidence to become stale.

RTL2GDSAGI is a Python controller that combines native tool execution with content-bound gate evidence. A model proposes a configuration change; deterministic code authorizes the changed surface and requires the affected native

checks before promotion. This paper tests three questions: whether these checks reject specific false-clean fixtures, whether live repairs survive native rechecking, and which implementation artifacts reproduce across runs.

The contributions are an executable candidate and gate contract, a native fault study that isolates six false-clean conditions, and a measured repair study with a comparator that accepts proposals without rechecking. The underlying synthesis, placement, routing, and verification algorithms come from the selected EDA tools. We claim detection of the tested fault classes; we do not claim that provenance alone establishes manufacturing qualification or that other controllers cannot implement comparable contracts.

2 Candidate and Gate Contract

2.1 Execution and content identity

The 21 stages are characterization, lint, simulation, constraint generation, synthesis, synthesis equivalence, pre-layout timing, floorplan, power-network generation, placement, clock-tree synthesis, post-CTS timing, routing, extraction, post-route timing, GDS stream-out, DRC, LVS, antenna, routed equivalence, and signoff. Pre-layout timing is advisory; post-CTS timing gates hold slack, and post-route timing gates both setup and hold. Signoff requires every declared hard obligation to be supported by positive evidence.

Verilator and Icarus Verilog provide lint and self-checking simulation. Yosys and EQY provide synthesis and equivalence [5]. OpenROAD provides physical implementation, extraction, and physical-stage timing. KLayout streams and verifies the layout. The artifact ledger records a producer, size, path, and SHA-256 digest for each output. Directory identities hash relative filenames and file contents. The candidate manifest additionally records configuration, constraints, selected PDK identity, and tool identities. The layer map and DRC/LVS decks are bound inputs because their contents change the meaning of the verification results.

Writing H for SHA-256, the candidate identity is

$$C = H(T, \{(k, H(A_k))\}_{k \in B}, P, V, H(K), H(I)) \,, \tag{1}$$

where T is the top, B the bound artifact keys, P the selected PDK identity, V the recorded tool identities, K the run configuration, and I the typed configuration state. The PDK identity includes its name, root, standard-cell library, and corner; it is not a hash of every file in the PDK installation. Verification decks and the stream-out map are hashed as members of B.

Each gate declares its consumed and produced artifact keys, report key, parser contract, and accepted tool identity. Release revalidation rehashes the files, checks gate lineage, and checks that DRC and LVS consumed the selected GDS. A certification identifier commits to the candidate identifier and gate records. The native KLayout invocation is also checked against the selected input and deck bytes, so an argument that selects another clean file is rejected. Figure 1 shows this boundary.

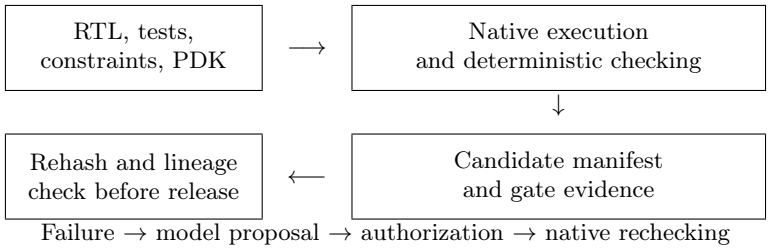


Fig. 1. The model proposes changes; the controller evaluates native evidence and candidate identity before promotion.

2.2 Required native evidence

DRC requires a usable report, the expected top, and the approved rule inventory under the exact deck identity. The selected runset has 211 normalized active categories; 213 raw declarations include two narrowly approved aliases. A zero-marker report with missing categories is insufficient. LVS requires agreement among exit status, log verdict, and a native database containing actual circuit cross-references. Its deck must contain an executable comparison and belong to the approved set. Stream-out checks nonempty geometry, resolved references, and layer numbers compatible with the PDK mapping. The PDN check requires supply nets, follow-pin rails, straps, and vertical via connections. These are structural obligations; IR drop and electromigration are outside the implemented contract.

Extraction compares the logical net identities in SPEF and routed DEF, rejects missing or duplicate nets, and requires complete parasitic sections. Standard punctuation escapes are decoded before comparing hierarchical names. Native signoff STA separately requires zero unannotated and partially unannotated drivers and finite setup and hold slack meeting the fixed guardband. The model cannot change clock period, uncertainty, output load, or verification decks.

2.3 Repair and equivalence

The Anthropic response schema distinguishes scalar configuration actions from exact RTL text patches. The safety checker limits writable fields according to the deterministic failure class and selects the earliest affected stage. A physical change invalidates downstream artifacts and evidence. Native rechecking must cover the action-derived obligations before the change becomes a verified repair; a model confidence value does not determine acceptance.

EQY partition cuts can make related outputs independent even when they feed joint next-state logic. The controller splits vector nets into scalar matches and groups the matched signals to prove their joint relation. Counter and shift-register controls pass this configuration; a control forcing a mapped counter-flop input to zero produces a counterexample. A later successful strategy resolves an

earlier unknown only for the same partition, while a counterexample or unresolved terminal result blocks release. This is a correction to proof decomposition, not a new equivalence algorithm.

3 Experimental Protocol

3.1 Designs and environment

The design set contains a register, registered multiplexer, eight-bit ALU, counter, bidirectional shift register, four-entry FIFO, and a 1,024-stage serial shift register. The latter supplies a larger synthetic case with 1024 mapped cells; it extends scale rather than adding an independent application domain. Each design has an authoritative self-checking testbench.

The PDK is SKY130A at Volare revision `bdc9412b3e46`. We use the HD cell library at its typical 25 °C, 1.8 V corner (`tt_025C_1v80`). The native container uses OpenROAD commit `41a51eaf4ca2` and KLayout 0.28.2; native pre-layout OpenSTA reports version 3.1.0. Baseline constraints use 10.0 ns, 0.25 ns uncertainty, and 0.05 pF output load. The stressed ALU uses 3 ns with the same uncertainty and load. Physical parameters and complete commands are retained in each run.

The paired baseline flows use source fingerprint `af31b39f6782`; the nine live repair episodes use the archived repair-cohort snapshot `18d948815ac6`. Both snapshots and exact commands are retained. The larger shift register uses two placement-padding sites, 25% core utilization, 0.35 placement density, an 80% conservative antenna prediction margin, eight native antenna repair iterations, eight routing workers, and passive filler cells. These settings change implementation and repair effort; the final antenna gate retains the original rule limits. Native antenna checking remains mandatory even when global-route repair predicts no violations.

We run each selected design twice from RTL through signoff. Concurrent jobs make elapsed time an observation of this environment, including container overhead, rather than an isolated performance comparison. A separate extractor validates candidate hashes, gate contracts, lineage, and certification identity before generating the tables. Cell area comes from the synthesis library report; die area comes from routed DEF coordinates and database units.

3.2 Native fault injection and comparators

The six fault classes in Table 3 use three replicates each. Stale-input trials run native DRC on a physically different, clean ALU layout with the same top name, while the candidate selects the current ALU layout. The rule-coverage trials disable FEOL checks in a copy of the real DRC deck. The LVS fixture writes a database-shaped file and prints a match without calling compare. The layer-map fixture performs a real stream-out with the mapping assignment omitted, then runs native DRC. PDN fixtures remove via connections from a retained DEF.

The final fixture substitutes a foreign GDS after a verified ALU repair while retaining the repair's DRC/LVS records.

The comparator accepts zero DRC markers after successful exit, an LVS match message after successful exit, or retained green status with a nonempty artifact for the cached PDN and post-repair cases. It deliberately omits coverage and candidate-identity checks. This controlled comparator isolates the verification obligation; it is not a measured implementation of OpenLane, ORFS-Agent, or another named external controller. Stale-input repetitions use the same two ALU candidates. Other fixtures span register, ALU, and FIFO candidates or three distinct foreign layouts, as recorded in the evidence.

3.3 Live repair study

Three repeats are declared for each of three injected failures: unpadded ALU spacing, an excessive register PDN strap offset of 50 μm, and ALU setup failure at 3 ns. Each episode uses Claude Sonnet 5 (Anthropic API; identifier `claude-sonnet-5`), automatic typed authorization, and the recorded retry policy. Every declared episode remains in the denominator. Model proposals can differ between repeats, and native tools determine the recheck outcomes.

A second comparator retrospectively replays the same authorized proposals, labeling each as promoted without repeating native checks. Native outcomes are used only to assess whether that promotion was supported. We evaluate this comparator at the proposal level, separately from the episode outcome. A native failed recheck establishes a confirmed false-clean promotion; an unavailable required recheck establishes an unsupported promotion. Earlier development and provider-transport failures are tabulated separately in the supplementary live-episode catalog.

4 Results

Table 1. Measured implementation results. Cell area is the synthesis-library area; die area comes from routed DEF. Timing uses extracted parasitics.

Design	Cells	Cell area (μm^2)	Die area (μm^2)	Setup slack (ns)	Hold slack (ns)
register	22	292.78	2393.17	6.954	1.204
mux4to1	48	480.46	3602.40	6.952	1.203
alu8	133	1018.48	6125.41	5.256	1.192
counter	47	416.65	3279.28	4.911	0.494
shift_reg	16	380.36	3091.92	6.892	0.554
fifo_buf	109	1721.65	9169.02	6.396	0.273
shift1024	1024	25624.58	115705.42	5.553	0.049

Table 2. Native verification and execution outcomes. Every design passed all 21 stages, both equivalence boundaries, LVS, and its authoritative testbench.

Design	Route/DRC violations	Antenna net/pin	SPEF nets	GDS (KiB)	Runtime (s)
register	0/0	0/0	35	131.8	32.0
mux4to1	0/0	0/0	87	162.0	47.6
alu8	0/0	0/0	157	498.2	75.3
counter	0/0	0/0	62	238.4	39.2
shift_reg	0/0	0/0	32	117.4	31.9
fifo_buf	0/0	0/0	126	486.6	54.7
shift1024	0/0	0/0	1210	3309.2	114.3

4.1 Implementation and verification

All 7 selected designs complete 21/21 stages with clean local certification. Table 1 reports mapped size, area, and extracted-parasitic timing. Detailed routing and DRC each report zero violations. Every DRC report contains the approved 211-category inventory. Native LVS matches; antenna checking reports zero violating nets and pins; and synthesis and routed equivalence pass. Every supplied testbench passes, and each SPEF covers every routed signal net counted by the identity check. Table 2 retains physical and runtime measurements.

4.2 False-clean evidence

Table 3. Native fault fixtures and the controlled status/count comparator. Replicates are defined in the protocol.

Fixture	Observed evidence	Naive clean	Blocked
Stale same-top GDS	DRC 0; input mismatch	3/3	3/3
Missing FEOL rules	DRC 0; 152 categories	3/3	3/3
LVS without compare	Match; zero cross-refs	3/3	3/3
Missing PDN connections	Nonempty DEF; zero vias	3/3	3/3
Missing layer map	DRC 0; invalid layer shapes	3/3	3/3
GDS swap after repair	Cached pass; digest mismatch	3/3	3/3

The comparator accepts 18/18 fault fixtures; the implemented checks reject 18/18. Disabling FEOL produces zero markers but only 152 normalized categories. Omitting the layer map also produces a native zero-marker result, while the stream-out check finds routing shapes on incompatible layer numbers. The fake LVS database has zero circuit cross-references despite its match message.

These results show why content identity, rule coverage, and comparison evidence are separate requirements. They establish detection of the six tested classes, not completeness against arbitrary checker or host compromise.

4.3 Live repair and timing stress

Table 4. All declared live repair episodes. Verified means the complete native signoff contract passed.

Injected failure	Trials	Verified	Escalated	Calls
Spacing: padding 0	3	3	0	3
PDN: offset 50 μm	3	3	0	3
Setup: period 3 ns	3	1	2	14

Of 9 declared episodes, 7 reach full verification (77.8%) and 2 escalate (22.2%). Spacing and PDN construction each succeed in three repeats; stressed timing succeeds in one of three. These are descriptive results on repeated fixtures, not a population repair-success estimate. The cohort makes 20 provider calls, with 174306 input and 50164 output tokens. Token counts are provider observations, not inferred cost estimates.

All three stressed ALU trials initially reach routed setup WNS -0.326192 ns under a generated 3.000 ns SDC period. In the successful trial, the delay-oriented mapping first remains negative; a subsequent placement update reaches 0.001586 ns with unchanged clock and protected RTL. The resulting layout passes all downstream checks. The small positive margin is a result at the stated typical corner and guardband, not evidence of multi-corner closure. A separate counter control at an unchanged 1.5 ns period reaches setup WNS -0.715689 ns and is blocked by the native timing gate. Figure 2 shows the measured ALU recheck sequence.

The comparator that skips rechecking promotes 20 authorized proposals. Native rechecks show 11 confirmed failures; 2 additional promotions lack a completed required target check. The production controller does not certify those proposal states. This result directly tests the danger of accepting a model's change as its own verification evidence. The earlier live episodes, including malformed responses and authentication/grammar failures, remain visible in the catalog.

4.4 Repeated artifacts and regressions

The repeat comparison in Table 5 uses content hashes. Routed DEF, mapped and routed netlists, and SPEF reproduce exactly for 6 design pairs. Timestamp-neutral GDS hashes reproduce for 6 pairs after zeroing only BGNLIB and BGNSTR date fields. All raw GDS hashes differ; some LVS serialization hashes also

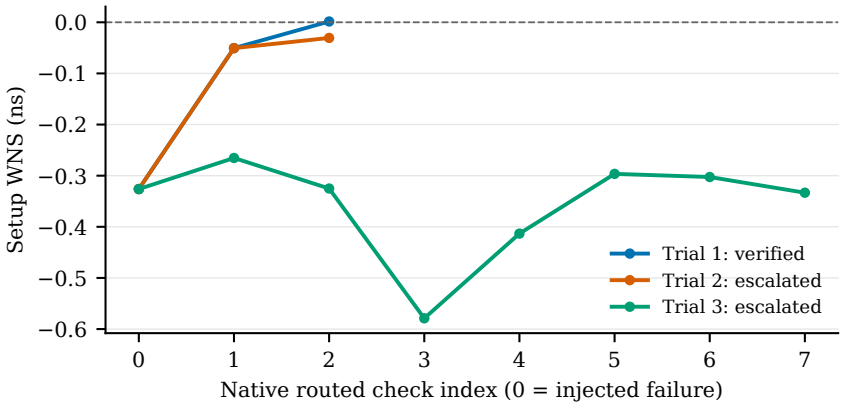


Fig. 2. Native routed setup checks at the unchanged 3 ns period. Lines end at the last completed routed check; the legend gives each episode's terminal outcome.

Table 5. Paired hashes. GDS (dates zeroed) changes only library/structure date payloads; full digests are supplied in the artifact.

Design	Routed DEF	SPEF	Raw GDS	Dates zeroed
register	Same	Same	Different	Same
mux4to1	Same	Same	Different	Same
alu8	Same	Same	Different	Same
counter	Same	Same	Different	Same
shift_reg	Same	Same	Different	Same
fifo_buf	Same	Same	Different	Same
shift1024	Different	Same	Different	Different

differ. The larger pair selects different native li1-to-met1 via variants, changing DEF and GDS bytes while mapped/routed netlists and SPEF remain identical. Eight routing workers were used for this case; the cause of the variant selection difference was not isolated. Candidate identifiers therefore identify actual artifact bytes rather than a normalized geometry class. Path-bearing characterization records can also differ between run directories. We claim reproducibility at the explicitly reported levels; full bitwise identity of every run file is not established.

The final regression suite reports 954 passes, zero failures and errors, and 15 skips. Skipped archived fixtures are not counted as passed verification. The supplementary record preserves incomplete scale probes including the 1,995-cell accumulator's spacing, antenna and diode-legalization failures and the ALU-bank antenna failures. These failed runs are excluded from the clean-design count.

5 Related Work

OpenROAD and OpenLane supply the implementation infrastructure [1,3]. METRICS2.1 provides structured flow measurement and tuning [2]; PROV provides general provenance relationships [4]. ChatEDA studies LLM-based task decomposition, script construction, and EDA execution [6]. ORFS-Agent studies iterative parameter optimization in OpenROAD-flow-scripts [7]. MCP4EDA connects language-model interaction to a complete tool flow and uses backend feedback for synthesis optimization [8]. FlowFixer-EDA describes log-grounded diagnosis, patch validation, and native tool feedback for repair [9]. These works establish that tool use and verified feedback already form part of LLM-assisted EDA research. EDAid uses multiple model-controlled agents to improve EDA workflow execution [12], extending the comparison beyond single-agent controllers. AuDoPEDA evaluates repository-grounded coding agents through executable OpenROAD changes and measured quality of results [13].

ChiBench supplies a curated RTL corpus for testing EDA tools [10]. DRVerify addresses physical-verification runset testing with known legal and violating layout structures [11]. Such work reinforces that a zero-marker count does not by itself establish rule intent or coverage. EQY supplies the equivalence engine used here [5]. The contribution of this study is the measured combination of native false-clean fixtures, candidate-content lineage, and action-derived rechecking. We make no unsupported priority claim that prior flows lack equivalent protections. A comparative claim against another controller would require a shared design set, tool environment, constraints, and verification obligations.

6 Discussion and Limitations

The fault fixtures demonstrate actual native false-pass outputs under the specified weak comparators. The guard rejects them through several mechanisms:

input identity for stale GDS, approved inventory for missing rules, native comparison evidence for fake LVS, structural connectivity for PDN, and layer mapping for stream-out. Hashes preserve identity under a trusted controller, host, tools, and PDK. They do not prove that an approved rule deck implements every foundry requirement or that tool code and gate records cannot be forged.

The repair study contains only three fixture types and three repeats per type. Its 7/9 outcome should not be generalized to unseen circuits or failure classes. The larger shift register is a regular synthetic circuit; full camera, processor, SRAM-macro, and multi-clock systems remain outside this evaluation. The declared local signoff omits multi-corner timing, on-chip variation, electrical power integrity, and foundry qualification. Formal equivalence establishes preservation of the RTL relation, while simulation checks the supplied behavioral tests rather than an external specification. The JSONL audit retains every accepted proposal, native recheck metric, and provider call count. Rollback can reuse a numbered attempt filename, so a complete raw prompt–response and intermediate-artifact archive for every proposal is not established. Final native artifacts and the cohort denominator are retained.

7 Conclusion

RTL2GDSAGI binds native verification evidence to a selected implementation and requires native rechecking of authorized model proposals. The measured flow completes 7 designs, including a 1024-cell synthetic shift register. All 18 native false-clean fixtures are blocked. Of 9 live repair episodes, 7 complete signoff and 2 escalate; the stressed ALU result preserves the 3 ns constraint. Repeated artifacts reproduce at the stated content and timestamp-normalized levels. The evidence supports the tested release contract and identifies the remaining limits on larger applications and verification coverage.

Code, data, and AI assistance

The public repository is `https://github.com/xp4t/rtl2gdsagi`. The versioned paper artifact is available at `https://github.com/xp4t/rtl2gdsagi/releases/tag/rtl2gdsagi-paper-20261012`. It includes source snapshots, exact commands, native reports, content digests, all retained live repair episodes, and scripts that regenerate the tables. The repair study uses Claude Sonnet 5 from Anthropic. Codex assisted writing and three final internal review passes addressing evidence, methods/citations, and final presentation. These passes share the same conversational context and are not independent external peer review.

A Repair Episodes and Incomplete Scale Probes

Table 6 lists each declared episode. Earlier live API episodes appear in Table 7; the supplemental JSON catalog maps each row to its retained run path and error

records. They are not pooled into the nine episodes because action schemas and provider transport changed during development. Interrupted scale probes, DRC failures, and antenna failures are retained with their measured stage outcomes and are excluded from the completed design count. Clock-buffer discovery, path handling, and terminal-verdict parsing are implementation prerequisites rather than independent contributions.

Table 6. The full declared repair denominator, with actual routed slack for the stressed setup trial.

Episode	Initial failure	Final evidence	Calls	Outcome
Spacing 1	1 marker	Full signoff	1	Verified
Spacing 2	1 marker	Full signoff	1	Verified
Spacing 3	1 marker	Full signoff	1	Verified
PDN 1	PDN-0185	Full signoff	1	Verified
PDN 2	PDN-0185	Full signoff	1	Verified
PDN 3	PDN-0185	Full signoff	1	Verified
Setup 1	-0.326192 ns	0.001586 ns	2	Verified
Setup 2	-0.326192 ns	Not certified	3	Escalated
Setup 3	-0.326192 ns	Not certified	9	Escalated

Table 7. All retained earlier live API episodes. API errors include authentication, transport and response validation. The catalog preserves their distinct protocols.

Case	First requested stage	Responses	Outcome
H1	drc	1	Verified
H2	drc	0	Stopped; API error
H3	drc	1	Verified
H4	drc	1	Verified
H5	sta_signoff	0	Stopped; API error
H6	sta_signoff	0	Stopped; API error
H7	sta_signoff	1	Stopped; action rejected
H8	sta_signoff	0	Stopped; API error
H9	lint	0	Stopped; API error
H10	lint	1	Stopped; API error

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